

THE JOINT LAW OF ZERO GAPS AND LOCAL MAXIMA OF HARDY'S Z -FUNCTION: A NUMERICAL STUDY AND A CUE EFFECTIVE-DIMENSION ANOMALY

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ABSTRACT. For consecutive ordinates $\gamma_n < \gamma_{n+1}$ of nontrivial zeros of the Riemann zeta function, let $\delta_n = \gamma_{n+1} - \gamma_n$ and let $M_n = \max_{\gamma_n \leq t \leq \gamma_{n+1}} |Z(t)|$, where Z is Hardy's function. The mean of M_n^2 is known: Conrey and Ghosh determined its leading order, and Hughes, Lugmayer and Pearce-Crump recently obtained the lower-order terms. The *joint* law of (δ_n, M_n) appears not to have been studied numerically. Using 3.4×10^5 gaps up to height $t = 1.6 \times 10^6$, we find that (i) the sample mean of M_n^2 matches the proven asymptotics to 0.05% (slope) and 1% (constant term), validating the dataset; (ii) the Pearson correlation of (unfolded) gap and maximum is $r \approx 0.75$ – 0.89 per window, far above the value 0.494 ± 0.007 attained by phase-randomized Gaussian surrogates with the same power spectrum ($+43\sigma$ on the $L = 9.86$ window); (iii) compared with characteristic polynomials of Haar-random unitary matrices (CUE), the zeta correlation at height t matches CUE of *effective dimension* $N_{\text{eff}} = L + c$, $L = \log \frac{t}{2\pi}$, with a metric-dependent shift: the rank-based (Spearman) effective dimension is *constant*, $N_{\text{eff}} = L + 2.0$, over $5.6 \leq L \leq 12.45$, while the Pearson-based one drifts from $L + 0.75$ to $L + 1.30$ (slope 0.098 ± 0.022 per unit L , 4.5σ); and (iv) the two effective dimensions differ by 0.93 ± 0.10 throughout, so the joint gap–maximum law of ζ is not that of any finite-dimensional CUE. A companion note traces both the split and the drift to transmitted prime waves.

1. INTRODUCTION

Let $Z(t)$ denote Hardy's function, so that $|Z(t)| = |\zeta(\frac{1}{2} + it)|$ and the real zeros of Z are the ordinates of the critical-line zeros of ζ . Write γ_n for the n -th positive ordinate, $\delta_n = \gamma_{n+1} - \gamma_n$ for the n -th gap, and

$$M_n = \max_{\gamma_n \leq t \leq \gamma_{n+1}} |Z(t)|$$

for the height of the wave between two consecutive zeros.

The *marginal* statistics of both quantities are classical objects. The gaps, after unfolding, follow GUE spacing statistics to remarkable accuracy (Montgomery [9], Odlyzko [10]); the maxima satisfy, under the Riemann Hypothesis,

$$(1) \quad \frac{1}{N(T)} \sum_{0 < \gamma_n \leq T} M_n^2 \sim \frac{e^2 - 5}{2} \log T,$$

by Conrey and Ghosh [2], with the full descending chain of lower-order terms obtained recently by Hughes, Lugmayer and Pearce-Crump [5]; see also [11] for the first moment with derivatives, and Milinovich [8] for bounds on higher moments.

The *joint* law of the pair (δ_n, M_n) is qualitatively constrained by folklore: Lehmer pairs [7] (very close zeros) force a very low intermediate maximum, and the large-gaps methods of

Conrey–Ghosh–Gonek implicitly exploit the converse. Quantitatively, however, we have not found in the literature: a measured correlation coefficient for (δ_n, M_n) ; a comparison of that correlation against spectrum-matched Gaussian surrogates; or a comparison against the CUE characteristic-polynomial model of Keating and Snaith [6]. This note supplies these three measurements and records a structural surprise in the third one.

Our effective-dimension framing follows Bogomolny, Bohigas, Leboeuf and Monastra [1], who showed that finite-height deviations of the *spacing distribution* of zeta zeros are captured by CUE of a well-chosen finite effective dimension. The observable studied here — the gap–maximum coupling — turns out to select its own effective dimension, and different projections of the joint law select *different* ones.

2. DATA AND VALIDATION

Two independent numerical engines were used.

Engine A (to $t \leq 7.5 \times 10^4$). Zeros from a standard table; M_n computed by bounded scalar minimization of $-|Z|$ per gap using arbitrary-precision evaluation of Z (mpmath), 10^5 gaps in total.

Engine B (windows up to $t = 1.6 \times 10^6$). A vectorized Riemann–Siegel implementation (main sum plus the C_0 correction), with zeros located by dense sign-change bracketing plus bisection, close pairs recovered by a targeted fine re-scan, and M_n from a 24-point interior grid with parabolic refinement. Against arbitrary-precision values the pointwise error is at most 7.6×10^{-6} over $10^5 \leq t \leq 1.6 \times 10^6$. Six windows of 4×10^4 gaps each were computed, centered at $t = 1.2, 2.0, 3.5, 6.0, 10, 16 \times 10^5$. In the overlap region the two engines agree to 10^{-4} in the gaps and to 0.1% (median: exact) in the maxima.

Both engines were validated against proven results. Writing $L = \log \frac{t}{2\pi}$ and fitting the windowed means of M_n^2 linearly in L ,

$$\widehat{\text{mean}}(M_n^2 | t) = aL + b,$$

we find $a = 1.1939$ (after correcting a measured 0.1% optimizer bias) against the Conrey–Ghosh constant $\frac{1}{2}(e^2 - 5) = 1.19453$, a deviation of -0.05% . For the constant term, differentiating the summatory asymptotic of [5] gives the local prediction

$$\text{mean}(M_n^2 | t) = AL + (\alpha_{-1} + 2A) + \frac{\alpha_0 + \alpha_{-1}}{L} + O(L^{-2}), \quad A = \frac{e^2 - 5}{2},$$

with $\alpha_{-1} = 5 - e^2 - 10\gamma_0 + 2e^2\gamma_0 = 0.368945$ and $\alpha_0 = -0.423285$ (Stieltjes constants γ_0, γ_1). This zero-free-parameter curve passes through all 26 of our windowed means within 2σ ; the fitted $b = 2.776$ deviates from the predicted effective value 2.749 by $+0.97\%$. We regard the dataset as anchored to two proven/published results.

3. THE CORRELATION AND TWO NULL MODELS

Within each window we unfold both variables using known scales: $\tilde{\delta}_n = \delta_n \cdot L/2\pi$ and $\tilde{M}_n = M_n/\sqrt{AL + 2.758 - 0.054/L}$, so that $\text{mean}(\tilde{\delta}) = 1$ and $\text{mean}(\tilde{M}^2) = 1$ to within 3×10^{-3} . Table 1 reports the windowed Pearson correlation r of $(\tilde{\delta}_n, \tilde{M}_n)$.

Null 1: spectrum-matched Gaussian surrogates. Following the surrogate-data method of Theiler et al. [12], one generates Gaussian processes with the same power spectrum as $Z(t)$ (random Fourier phases), extracts their zeros and between-zero maxima, and computes the same correlation. On the $L = 9.86$ window (4×10^4 gaps; zeta and surrogates extracted

TABLE 1. Windowed correlation of unfolded gap and maximum, and the effective CUE dimension shift (Section 4).

L	n (gaps)	r (Pearson)	$N_{\text{eff}}^P - L$	$N_{\text{eff}}^S - L$
5.60	1 029	0.8917	+0.75	+1.70
6.30	2 322	0.8735	+0.76	+1.80
6.99	5 174	0.8566	+0.77	+1.81
7.69	11 418	0.8394	+0.87	+1.93
8.39	24 988	0.8238	+0.95	+1.98
9.08	54 299	0.8096	+0.99	+2.00
9.86	39 992	0.7947	+1.04	+2.00
10.37	39 997	0.7849	+1.13	+2.01
10.93	39 998	0.7754	+1.20	+2.01
11.47	39 998	0.7663	+1.31	+2.14
11.98	39 999	0.7589	+1.31	+2.07
12.45	40 000	0.7521	+1.30	+2.10

identically from the same fine grid), 40 phase-randomized surrogates give $r_{\text{null}} = 0.494 \pm 0.007$ against the zeta value 0.794: a $+43\sigma$ excess. (An earlier small-scale probe over $t \in [50, 500]$, ≈ 260 gaps, found the same null level at $+6\sigma$.) The gap–maximum coupling of ζ is therefore far stronger than that of *any* Gaussian process with its spectrum; it depends on structure beyond second-order statistics.

Null 2: CUE characteristic polynomials. For an $N \times N$ Haar-random unitary U with eigenphases ϕ_j , put $|\Lambda(\theta)| = \prod_j 2 |\sin \frac{\theta - \phi_j}{2}|$, and for each of the N cyclic eigenphase gaps record the pair (gap, interior maximum of $|\Lambda|$). The density matching $N \leftrightarrow L$ is the standard one of [6]. Table 2 reports the correlation for $N = 5, \dots, 20$ (1.5×10^6 gaps per N ; Monte-Carlo uncertainties, estimated from ten independent batches per N , are below 6×10^{-4}). We are not aware of a published computation of this per-gap correlation; its closest published relative is the demonstration by Fyodorov, Gnutzmann and Keating [3] that the *global* maximum of $|\Lambda|$ preferentially sits in large spectral gaps.

TABLE 2. Gap–maximum correlation for CUE characteristic polynomials.

N	Pearson	Spearman	N	Pearson	Spearman
5	0.9293	0.9683	13	0.7632	0.8860
6	0.9011	0.9545	14	0.7486	0.8793
7	0.8750	0.9420	15	0.7365	0.8729
8	0.8512	0.9303	16	0.7258	0.8671
9	0.8305	0.9198	17	0.7130	0.8612
10	0.8109	0.9103	18	0.7022	0.8560
11	0.7929	0.9014	19	0.6931	0.8512
12	0.7773	0.8933	20	0.6825	0.8466

At the conventional matching $N = L$ the zeta values sit consistently *below* CUE: $r_\zeta - r_{\text{CUE}}(N=L) = -0.019$, essentially flat across all windows. Thus ζ is bracketed by the two

nulls,

$$\underbrace{0.505}_{\text{Gaussian, same spectrum}} \ll \underbrace{r_\zeta \approx 0.82}_{(\text{at } L \approx 8.4)} < \underbrace{0.841}_{\text{CUE, } N=L},$$

strongly coupled like a random-matrix model, but measurably less than one.

4. AN EFFECTIVE-DIMENSION ANOMALY

To express the deficit structurally, define $N_{\text{eff}}(L)$ by $r_{\text{CUE}}(N_{\text{eff}}) = r_\zeta(L)$. We deliberately fit *no* functional form to $r_{\text{CUE}}(N)$: an earlier version of this analysis used a quadratic in $1/N$, which is rejected against the Monte-Carlo errors of Table 2 and, as an adversarial re-analysis showed, manufactured a spuriously constant Pearson shift. Instead we invert a shape-preserving monotone (PCHIP) interpolation of the precise integer- N values, and propagate both the zeta-side sampling errors and the CUE Monte-Carlo errors through the local slope. The last two columns of Table 1 give the result.

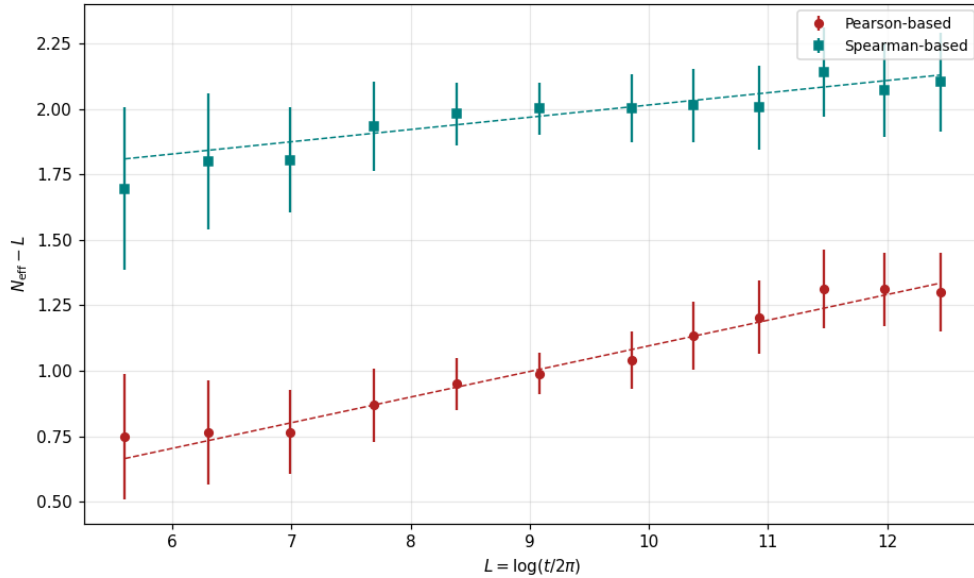


FIGURE 1. The effective-dimension shift $N_{\text{eff}} - L$ across all twelve windows. The Spearman-based shift is constant near +2.0; the Pearson-based shift rises with L . Dashed lines: weighted linear fits.

Observation 1. Over $5.6 \leq L \leq 12.45$ (heights t from $\sim 1.2 \times 10^3$ to 1.6×10^6), the rank-based shift is constant,

$N_{\text{eff}}^{\text{Spearman}} - L = +1.99$ (constant-model $\chi^2/\text{dof} = 0.34$; a linear trend improves χ^2 by only 3.0), while the Pearson-based shift is not: it rises from +0.75 to +1.30, with weighted linear slope $+0.098 \pm 0.022$ per unit L (4.5σ ; constant-model $\chi^2/\text{dof} = 1.89$).

Observation 2. No single effective dimension accounts for both statistics: the Pearson and Spearman projections of the joint law $(\tilde{\delta}, \tilde{M})$ require shifts differing by 0.93 ± 0.10 , stably across all windows. Consequently the joint gap-maximum law of ζ at finite height is not that of any finite-dimensional CUE.

Three remarks. First, the pattern — a constant rank-based shift and a drifting amplitude-sensitive one — points at the prime-oscillation terms of the explicit formula: they modulate the *amplitudes* on scales longer than one gap, so a rank statistic of the ordering is largely immune while the Pearson statistic absorbs their L -dependent transmitted variance. The companion note [4] measures exactly this mechanism and shows the two statistics coincide once the measured prime waves are stripped. Second, the observable-dependence of N_{eff} is consistent with the picture of [1], where the spacing-shape observable selects the very different effective dimension $N_{\text{eff}} = L/\sqrt{12\Lambda}$, $\Lambda = 1.57314\dots$; each statistic probes the arithmetic corrections through its own projection. The two effective dimensions are not the same quantity and differ numerically by a factor of about five over our range (at $L = 11.98$: $L/\sqrt{12\Lambda} = 2.76$ versus the correlation-calibrated $N_{\text{eff}} = 13.4$); ours is defined only through the r -statistic. Third, we deliberately refrain from proposing closed forms for the constants 2.0 and 0.93: in an earlier phase of this project two closed-form candidates for a related slope observable survived to $T = 6 \times 10^3$ and $T = 10^4$ respectively and were then cleanly refuted at larger heights; indeed the “constant Pearson shift” reported by an earlier draft of this very note fell the same way, to an adversarial re-analysis of the interpolation.

5. OPEN PROBLEMS

(1) Derive the local mean of M_n^2 *conditioned on the gap* — equivalently the regression slope behind r — from the methods of [2, 5]; the machinery (the auxiliary function Z_1) sums over extrema and may admit a gap-weighted variant. (2) Derive the constant rank shift 2.0, the Pearson drift, and the split 0.93 from the explicit formula, in the spirit of [1]; the companion note [4] supplies the measured prime-wave inputs. (3) Compute $r_{\text{CUE}}(N)$ analytically; even the $N \rightarrow \infty$ limit (the sine-process value) appears to be unknown. (4) Recover the discriminating power of the N_{eff} test at larger heights (at the 10^{12} -th zero, $t \approx 2.7 \times 10^{11}$, 10^4 gaps leave an uncertainty of ≈ 0.8 on N_{eff}), which needs $\gtrsim 10^5$ consecutive zeros there.

REPRODUCIBILITY

All code (analysis scripts 27–60), the datasets of gap–maximum pairs, and a dated research log are available in the repository <https://github.com/azizran/riemann-zero-statistics> (directory `qm_riemann/`), archived at Zenodo, doi:10.5281/zenodo.22942475.

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